

DELIVERABLE # 2

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October 5, 2025

CENG-322-OCC, Software Project/Hardware Production

FINAL DOCUMENT LAYOUT

Please submit a single PDF document per group. This means that the document should be professionally organized and have a uniform style throughout. It should look as though it came from one team, not 4 separate students. Please note that instructors may choose to run your submission through TurnItIn or compare the submission with other students from other sections for academic integrity purposes.

A single PDF document with the following sections denoted using page numbers, headers/footers and a table of contents:

All team members must present, missing members will be assigned 0.

Must follow the sequence below:

Use proper formatting

1. Document Title: **Deliverable 2**

2. Team Name: **LIFE (*Light, Indoor air, Freshness, and Energy*)**

3. Project Name: **LIFE Environmental Solutions**

4. Group Number: **11**

5. Student's names and IDs: **Mohamed Ali Student ID: N01440760, Jonathan Akabuike Student ID: N01510573, Kieran Sharma Student ID: N01548225, Farhan Habibzai Student ID: N01610299**

6. Table of Contents(Along with team contract)

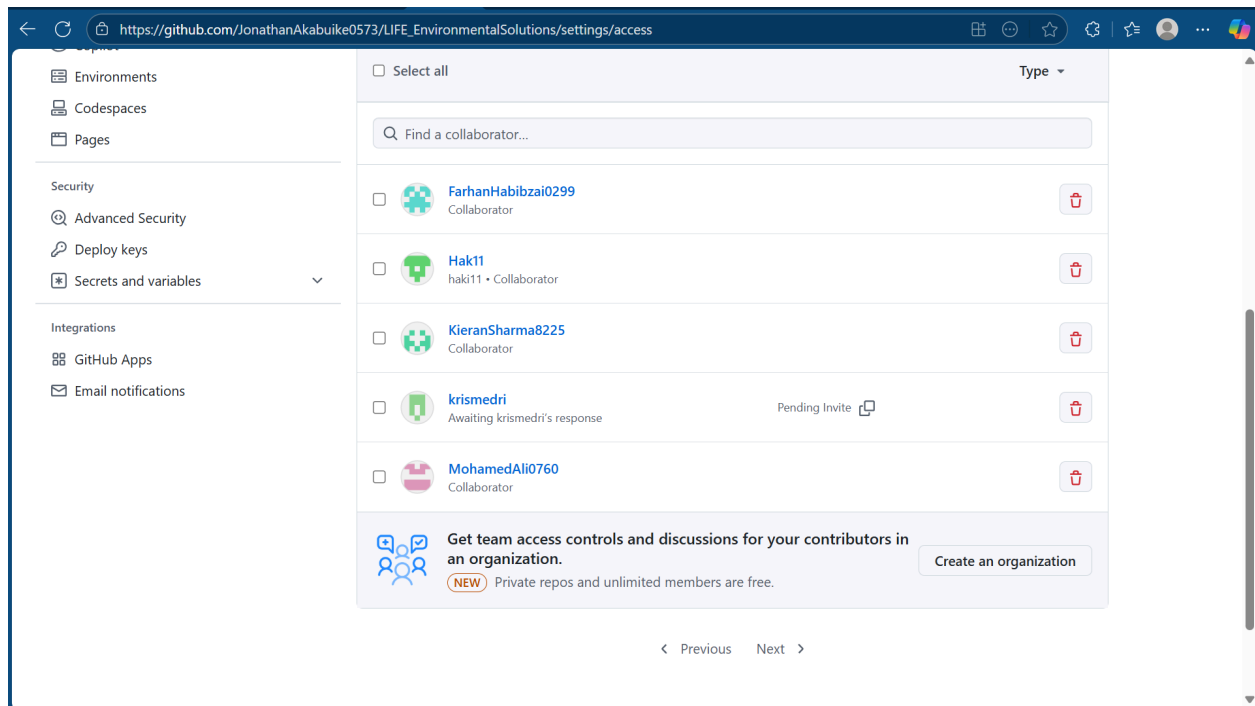
7. Table below with signatures

8. Members Info and Participation. Effort was put on the deliverable (0 – 100%), students will be marked based on their effort, i.e. if the team gets 8, and one student effort was only 50%, that student mark will be adjusted to $8 * 50\% = 4$

Name	Student ID	Github ID	Signature	Effort
Mohamed Ali	n01440760	MohamedAli0760	M.A.A.	100%
Jonathan Akabuike	n01510573	JonathanAkabuike0573	J.A.	100%
Kieran Sharma	n01548225	KieranSharma8225	K.S.	100%
Farhan Habibzai	n01610299	FarhanHabibzai0299	F.H	100%

1. In the pdf, write the requirement and comment on it for each item. Include the question and the answer.

2. Project Scope and goals that are targeted in this course.: **The project scope is to make the software and the hardware intertwined and to have the app running showing different readings and to have a clear showing of air quality in the system.**
3. GitHub Repo link and strategy. All members must contribute to the repo. Explain the github strategy.: **Our github strategy is to combine all of our work on single branches, pull from each other, then merge everything into the master branch. We also for github make sure that the merge is successful for the master branch and that the running of the app doesn't crash. We make sure we don't work on the master branch and we make sure we don't have any conflicts on each other branch by making sure we communicate whenever we commit and push, we pull everything and then we can commit and push. That way it is safe and we can assure no problems will occur.**
4. Verify the link is working.
5. Screenshot showing you have invited your hardware prof to the repo (explain to your hardware prof, the process to accept your invitation).

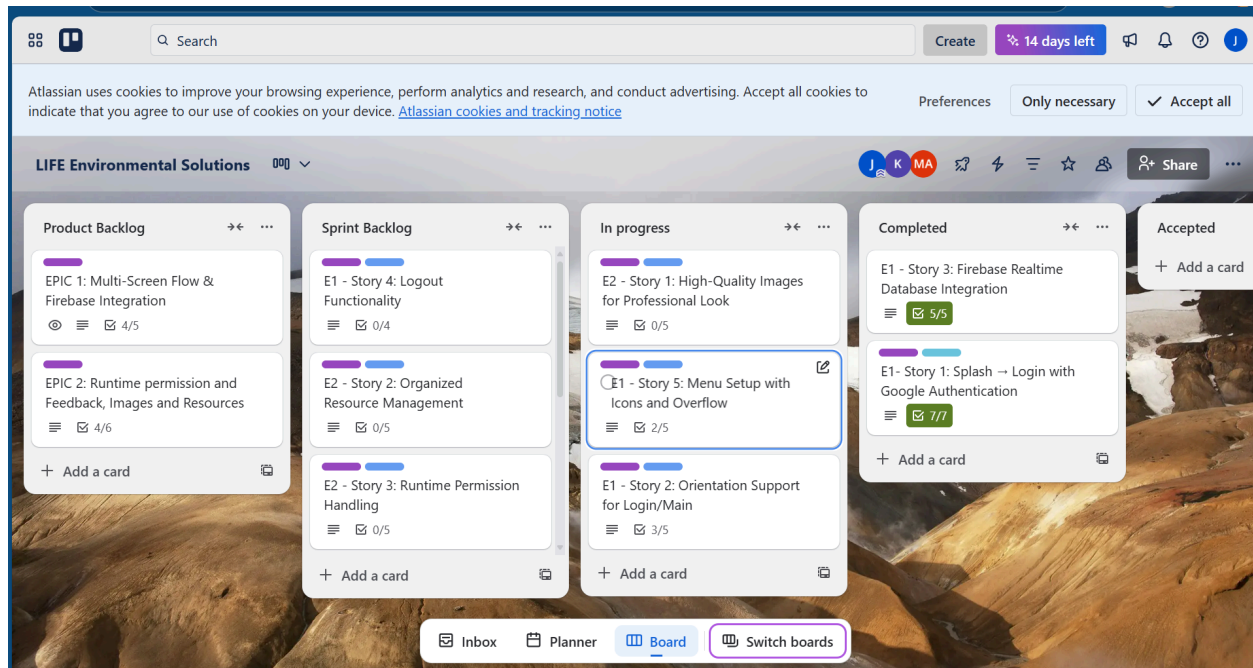


6. Each member must have a minimum of 5 commits.
7. Marks deducted for members who are not meeting the minimum of 5 commits (i.e. if only 4 commits, 20% percent deducted, ...etc.).
8. Minimum of 5 stories and each story must be split to minimum of 4 tasks. Use one of the online tools to build the stories and tasks, i.e. use Trello (<https://trello.com/en>) or any other appropriate tool:

9. Take a screenshot showing clearly the stories and breakdown of tasks with owner, status, start/end date, size, and priority.

10. Refer to lecture 4 on how to build a dashboard for the stories and tasks.

: **We used Trello to create a scrum dashboard showing the Stories and Task**



11. Tasks which are marked done, explain how did you meet the DoD criteria. (Android Studio)

12. List of items you included in the DoD. (Android Studio)

13. Business Model Canvas, all the 9 fields are clearly shown and explained.

14. Create Gantt chart showing main milestones and work progress. You can use <https://try.airtable.com/gantt2> or any other tool. Do the research and select the proper tool. Excel or word will not be accepted.

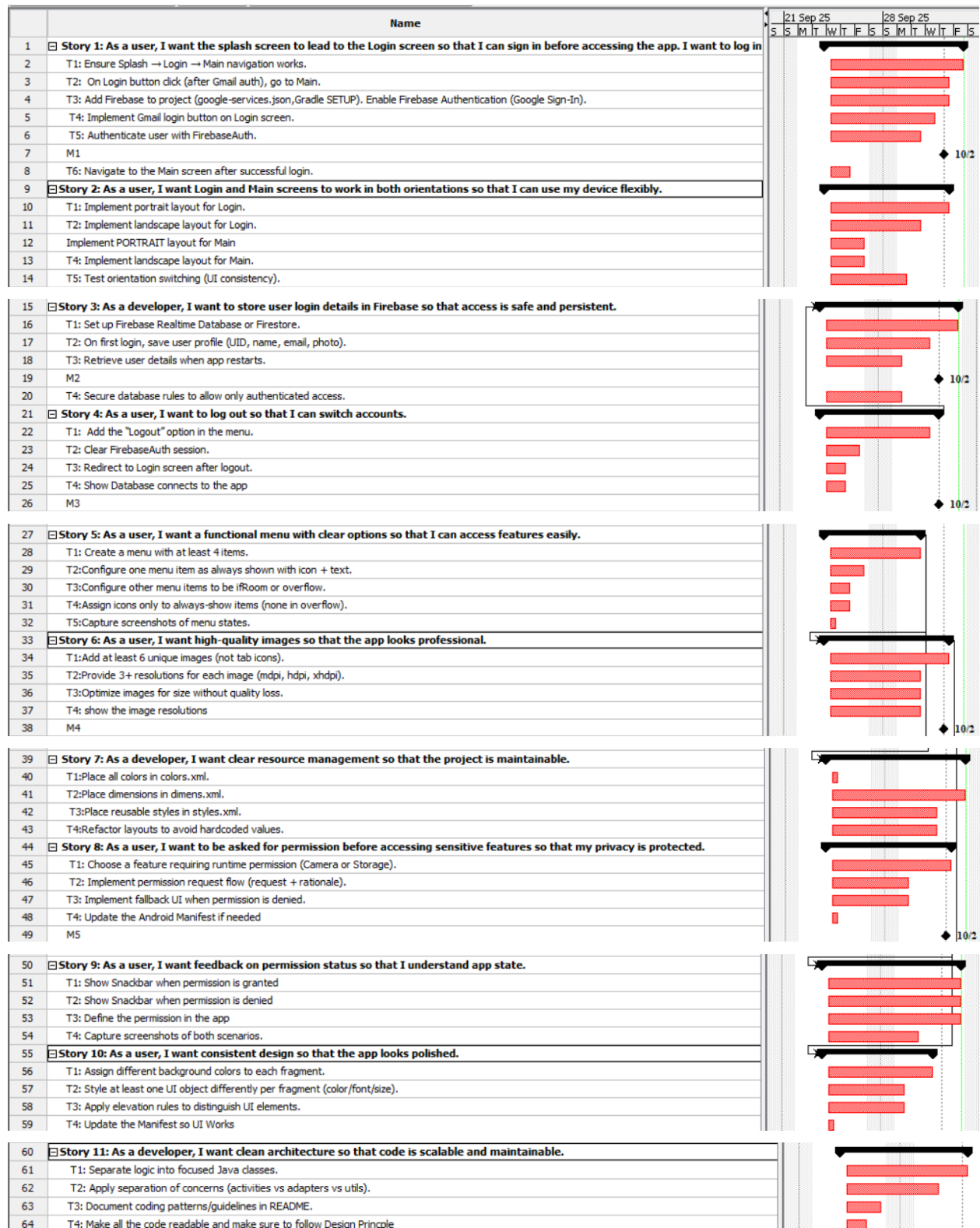
15. The Gantt chart must show the work for a minimum of 10 components(stories) and the time lines. Focus on main components and not every small task.

16. The Gantt chart should show the project from start to end.

17. Refer to Week 5 material, on what questions the Gantt chart answers, i.e. project timeline, assignee, ...etc.

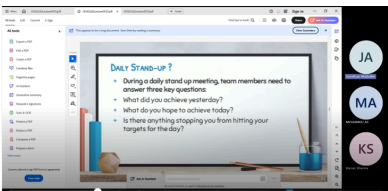
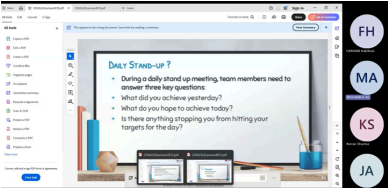
18. Gantt chart is different than your Scrum Dashboard, and does not show the small task's details.

19. Multiple screenshots of the Gantt charts showing the Details.



20. Screenshot showing you have created the DB on the cloud, with some details and type of DB used. (We have chosen Firebase Database)

21. Explain how you are planning to use the DB.: we plan on using the DB by showing it
22. Coding work progress since deliverable 1. What additional features/functionality added since deliverable 1.: **We added in pictures to show what our sensors will look like, we added in colors and different styles, as well as different resolutions to our images.**
23. A Table records the daily stand-ups outcome, use any tool you like, include a screenshot into the document. Table, include date and info. Show minimum of 3 meetings.

Meeting 1	link: Meeting in General-20250930_223654-Meeting Recording.mp4	
Meeting 2	Link: Meeting in General-20251003_232146-Meeting Recording.mp4	
Meeting 3	Link:	Meeting held on October 2, 2025

24. Document what design principle you used, copy the code and comment on it.
25. Create a subfolder called deliverable2 under docs in the repo and add the pdf file.
- If document not in pdf, or not the proper name (marks will be deducted!)
- Please save the PDF document, i.e.
- TeamName_ProjectName_GroupNumber_2.pdf
- One member to submit the pdf file.

- Presentation

Presentation Material

- 1- **Android Studio with the app running and simulator or real device.**
- 2- Power Point Document highlights the main components. Include Team Name, Project Name, Group Number, Student Names and their student Ids.

- 3- Demo of the app.
- 4- Screens and flows.
- 5- Business Model Canvas.
- 6- Sprint dashboard.
- 7- Github and how it was used.
- 8- Gantt chart summary.
- 9- All members must present. Zero for members who miss the presentation.
- 10- Practice on presenting, and be ready, each team member will have 2 - 3 minutes to present.
- 11- Commit and push the ppt document into deliverable 2 Folder.

-- Android Studio

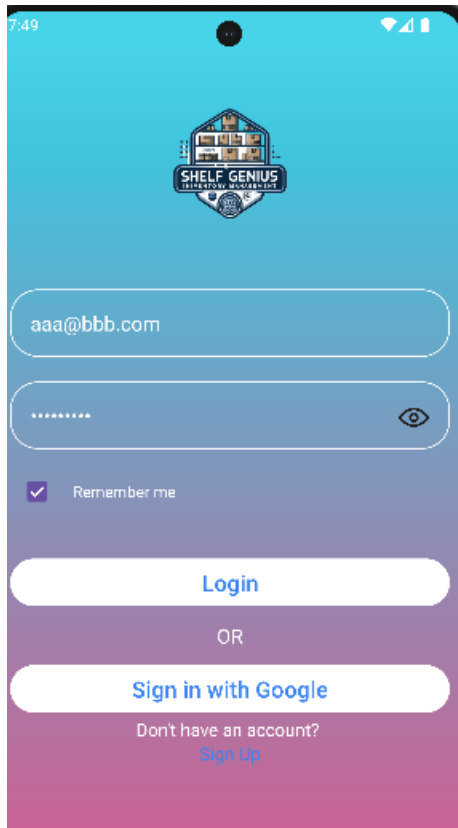
Deliverable 2:

Project pushed into in github:

- 1- Minimum API 30.
- 2- Target API 36/37.
- 3- Use a Navigation Drawer in combination with another layout. (Navigation drawer + bottom navigation)
- 4- Design of all the screens in the app. Complete screens of the app without functionality. 2 for each member
- 5- Design of the settings screen without functionality, should clearly show what a user can set/modify in the settings screen. Must include a minimum of 4 features, one of them is lock the screen to portrait.
- 6- Implement only one functionality in the Settings Screen.
- 7- Implement Splash Screen when user clicks on the homescreen icon, and then land the user on the Main Screen.
 - a. Use implementation `libs.core.splashscreen`
 - b. Splash screen appears for 3 seconds, then user lands on the main screen.
 - c. Design of two splash screens, one for default theme and one for the night theme.
 Implement some type of animation on the splash screen.
https://www.youtube.com/watch?v=eFZmMSm1G1c&ab_channel=PhilippLackner
- 8- All screens must support English and French.
- 9- Must have a minimum of 5 screens (including the login screen).

More screens to be added in the coming deliverables.

10- User landed from Splash screen into Login Screen, then when clicking on Login, it lands on the Main Screen.



11- Screens must include a total of minimum 6 images. All images must have minimum of 3 resolutions.

12- Use proper images for the tabs, not counted as of the 6 images.

13- Login and main screens must have landscape and portrait layouts.

- 14- If you can not meet minimum screen or image requirements, need to think about your project scope and expand the functionality. Identify the expanded functionality.
 - 15- The app must have a menu, screenshots must show the menu. Must have a minimum of 4 options (whether always, ifRoom or never will depend on the functionality).
 - 16- Menu items which are shown on the menu bar should have proper images. Menu items in the overflow should not have an image.
 - 17- Must have one menu item set to always with image and Text.
 - 18- Implement at least one functionality that requires runtime permissions. Asks for the proper permission. Display proper screen if permission not granted.
 - 19- Display a snackbar whether permission is granted or denied.
 - 20- Back key dialog in deliverable 1 must have a proper image. (DONE)
 - 21- To follow design patterns, have your Java classes focus on certain functionality. Do not make it too big, and hard to maintain.
 - 22- Follow guideline to use proper elevation for different UI elements.
 - 23- Select different background colors for the fragments. Each fragment must have a different background color.
 - 24- Each fragment must have at least one object with a different style than the default style (i.e. color, type, fontType, fontSize,...etc).
 - 25- Demonstrate the use of styles in objects.
 - 26- Demonstrate the use of colors.xml, dimens.xml, and styles.xml. Avoid hardcoding any of the styles.
 - 27- Plus deliverable 1.
 - 28- Make sure all the work is merged and pushed into main branch in github. Only the main branch will be marked. Other branches are for your own exploration.
 - 29- Create a subfolder called deliverable2 under docs in the repo and add the pdf file.
 - 30- Prof should be able to clone and run your app without any issue.
 - 31- App must not have compilation errors.
- Deliverable 1:
1. Refer to deliverable 1 folder.

